

AI for Bharat Hackathon

Powered by **aws**



Team Name : API Avengers

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Problem Statement : Complex procedures, language barriers, and costly legal aid deny basic justice to millions in Bharat.

Brief about the Idea:

- Nyaya Mitra AI is a multilingual, voice-first legal assistant built for Bharat.
- Democratizes access to justice using natural conversational AI on simple smartphones or IVR.
- Acts as a "Virtual Justice Friend" to educate citizens on their rights and government schemes.
- Proactively drafts official complaints, RTIs, and FIRs in plain local languages.
- Bridges the immense gap between complex legal/judicial systems and ordinary citizens.

Your solution should be able to explain the following:

- How it solves: Translates intimidating "legalese" into simple local dialects to empower users instantly.
- Differentiation: Goes beyond generic advice to actually draft formatted, ready-to-submit legal documents.
- USP (Zero Middlemen): Eliminates reliance on corrupt agents by guiding citizens directly through official portals.
- USP (Voice-First): Breaks the literacy barrier entirely with native speech-to-speech interaction.
- USP (Mediation Focus): Proactively suggests alternative dispute resolution to save time and court resources.

List of features offered by the solution

- **Multilingual Voice Bot:** Conversational interface supporting native languages (Hindi, Marathi, Tamil, etc.).
- **Legal Knowledge RAG:** Highly accurate answers grounded strictly in verified government acts and FAQs.
- **Automated Document Drafter:** Auto-generates PDFs or SMS texts for official complaints based on user stories.
- **Legal Notice Scanner (OCR):** Simplifies physical court summons via smart camera scanning and summarization.
- **Smart Escalation:** Direct hand-off protocol to human free legal aid (NALSA/DLSA) for critical situations.

Process Flow diagram or Use-case diagram

Nyaya Mitra AI – AI-Powered Legal Assistant on AWS

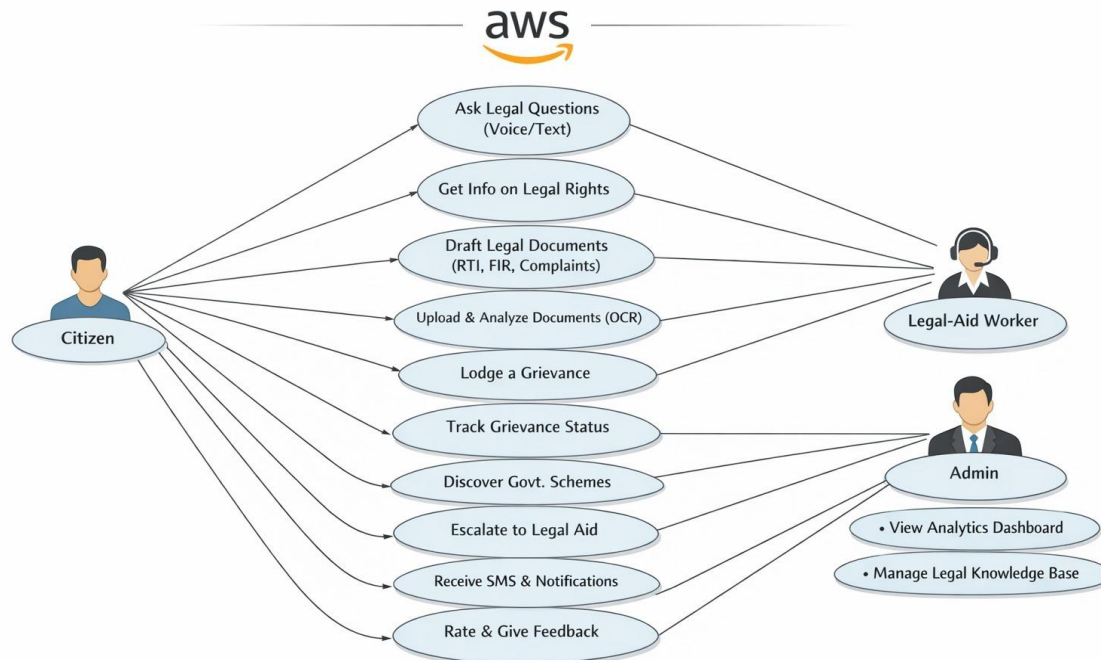


Figure No. 1.1 (Use-Case Diagram)

Wireframes/Mock diagrams of the proposed solution



Figure No. 2.1 (Home Screen)



Figure No. 2.2 (Chat Interface)



Figure No. 2.3 (Document Scanner)

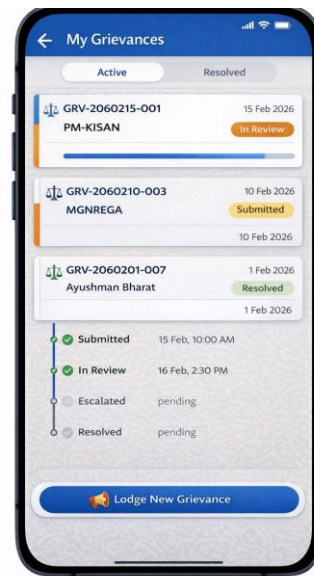


Figure No. 2.4 (Grievance Tracker)

Architecture diagram of the proposed solution:

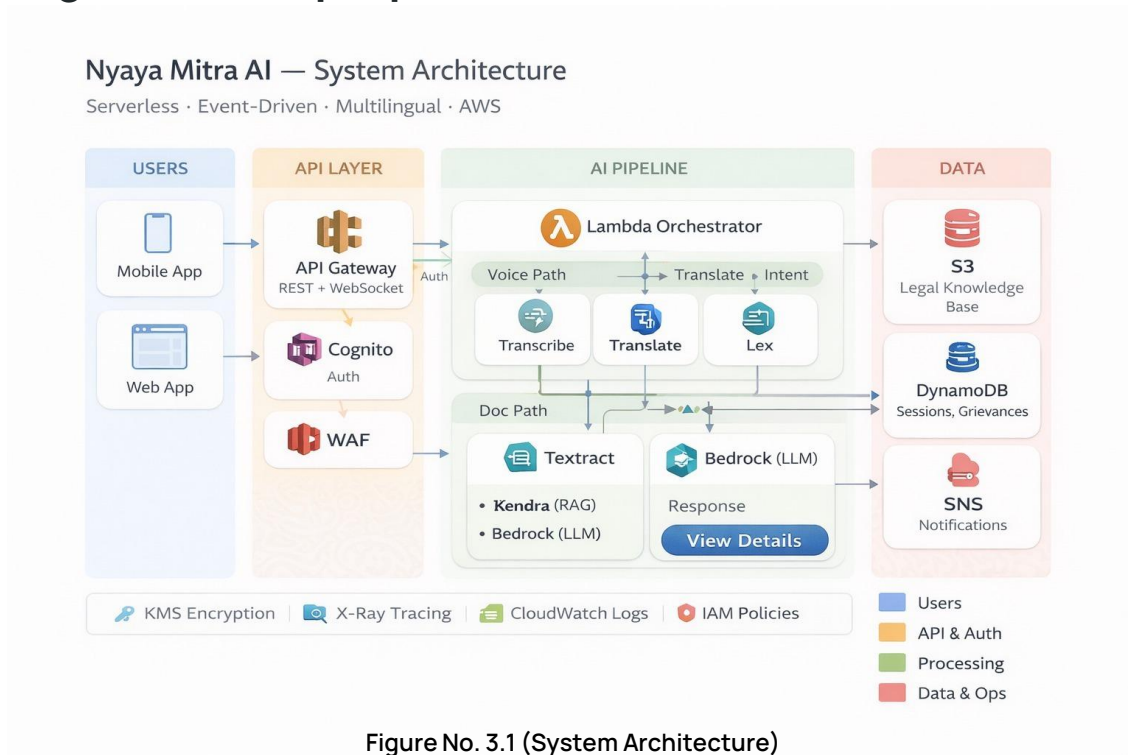


Figure No. 3.1 (System Architecture)

Technologies to be used in the solution:

- Frontend: React Native for mobile accessibility and Next.js for the web platform.
- Core AI & Reasoning: Amazon Bedrock (LLM) for legal logic and Amazon Kendra for RAG search.
- Conversational Interface: Amazon Lex for multi-turn intent classification and dialogue flow.
- Multilingual Voice Pipeline: Amazon Transcribe, Translate, and Polly to break language barriers.
- Document Processing: Amazon Textract for OCR to scan and summarize physical court notices.
- Backend & Database: AWS Lambda for serverless orchestration and Amazon DynamoDB for session state.

Estimated implementation cost :

- **Serverless Efficiency:** 100% pay-per-request execution (AWS Lambda) keeps idle running costs at absolute zero.
- **Low MVP Investment:** Using managed AWS AI services avoids expensive, from-scratch model training costs.
- **Pay-as-you-go Database:** Amazon DynamoDB ensures highly cost-effective data storage during the prototype phase.
- **Open-Source Frontend:** React Native and Next.js eliminate any UI licensing or heavy development expenses.
- **Scalable Economics:** The architecture ensures that infrastructure costs only scale linearly with actual user adoption and impact.

